

Biraj Shrestha
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Professional Summary:

Computer Science graduate with hands-on experience in software development and databases. Proficient in OOP, algorithms, data structures, and agile teamwork. Comfortable with Unix and enterprise systems. Quick learner and problem-solver, eager to contribute across the software development lifecycle and build impactful solutions.

Education:

Youngstown State University

Bachelor of Science in Computer Science

Minor: Mathematics, Data Analytics Certificate

GPA: 3.72/4.0, Dean's List, President's List

YSU Honors College Scholarship, YSU Presidents International Award

Youngstown, OH

Graduation: December 2025

Technical Skills:

Programming Language: C++ and Java

Web Development: HTML, CSS, JavaScript, Node.js

Database: SQL, MYSQL, MongoDB

Data Analysis: R, Tableau

Applications: Git, GitHub, WinSCP, TortoiseGit, PL/SQL Developer, Azure, Team Dynamics

OS: Windows, Linux, Mac

Projects:

VDI File Handler

- Implemented C++ file copier for large Virtual Disk Image (VDI) files using low-level file I/O.
- Developed functions to read, write, and manage disk blocks for efficient file handling.
- Applied operating system concepts such as inodes, block allocation, and buffer management.
- Ensured data integrity by validating copy operations and handling edge cases.

Sudoku Solver

- Built a Sudoku solver in C++ using Breadth-First Search (BFS) to efficiently fill a 9x9 grid.
- Designed constraint validation checks for rows, columns, and 3x3 subgrids to ensure correctness.
- Implemented a console-based visualization for real-time tracking of the solving process.

Senior Capstone: Ontology Based Event Modelling

- Designed a structured domain model relationship representing events, actors, location and interactions using conceptual graphs in CoGui with Nepal Government as case study.
- Modeled entity relationship with focus on clear system structure and data consistency.
- Documented design decision, assumptions and implementation limitations, identifying future extensibility opportunity.

Work Experience:

Student Programmer 1, Youngstown State University, Youngstown, OH

(Fall 2024- Summer 2025)

- Developed and maintained applications supporting the University's ERP (Banner) system
- Wrote, tested and documented code for system integrations, reports, and custom applications
- Addressed users requests by modifying reports and optimizing existing solutions to improve usability and reliability.

Student Field Technician, Youngstown State University, Youngstown, OH

(Fall 2023- Fall 2024)

- Resolved urgent classroom technology issues and maintained printers and devices
- Installed, imaged, and configured computers using Altiris; set up labs with modern equipment
- Maintained printers, AV equipment's, and endpoint devices across campus.

Student Technician, Youngstown State University, Youngstown, OH

(Fall 2022-Fall 2023)

- Provided support for LMS and portal applications, creating and managing service tickets
- Diagnosed and resolved system and software issues across multiple device types
- Responded to user inquiries via phone and ticketing systems in a fast-paced environment.

Activities:

Member, Competitive Programming Club

International Representative SGA, YSU, OH

Communication Director for NSA, YSU,